

K2-30b

R-2263

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-30_221112 · created 2026-08-16T01:34:44+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459895.872 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.043
Transit depth (Rp/Rs)^2	1.92 +/- 1.19 [%]
Orbital Inclination (inc)	84.98 +/- 2.32
Airmass coefficient 1 (a1)	0.964 +/- 0.022
Airmass coefficient 2 (a2)	0.033 +/- 0.033
Scatter in the residuals of the lightcurve fit is	2.28 %
Best Comparison Star	#3 - [380, 358]
Optimal Aperture	3.41
Optimal Annulus	10.81
Transit Duration (day)	0.046 +/- 0.043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.043 vs 0.1266 (deviation 0.29σ)
Duration fit vs published	0.046 d vs 0.0971 d (deviation 1.19σ)
Mid-time O–C	24.6 min
Depth SNR	3.2
Residual scatter	2.28 %

Light curve

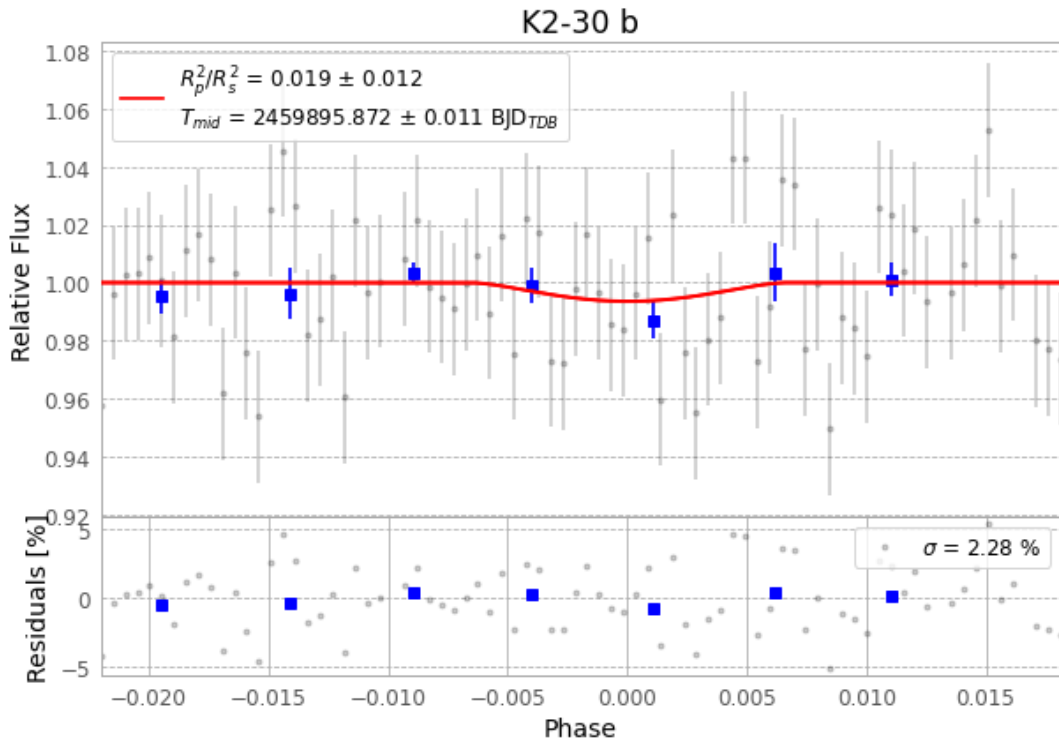


Figure 1 — FinalLightCurve_K2-30 b_2022-11-11.png (SHA-256 89289d81ccbc5f0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 192], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 2f034671b6f3dff5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

80 FITS frames (53 MB), K2-30221112063615.FITS → K2-30221112103315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-30-221112.log (c3ddab20939f...), FinalLightCurve_K2-30 b_2022-11-11.png (89289d81ccbc...), FinalLightCurve_K2-30 b_2022-11-11.csv (756cca888471...)

Compute resources

Wall time 112.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 01:34Z.